



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of
Built Environment and Surveying

FACULTY OF ALAM BINA & UKUR

SEMESTER 1/20252026

SBEQ 1343 – INTRODUCTION TO QUANTITY SURVEYING

SECTION EB1 (Group 2)

GROUP ASSIGNMENT PHASE 1:

CASE STUDY

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1.0 Introduction

In the construction industry, quantity surveying is essential because it guarantees that projects are planned, managed, and finished within the allocated budget while meeting the required time and quality goals (Ashworth et al., 2013). Throughout the whole project lifecycle, from the first feasibility stage to the final account settlement, a quantity surveyor is in charge of cost management, contract administration, procurement advice, and financial control. Their participation is essential to reducing financial risks and guaranteeing clients receive value for their money (Royal Institution of Chartered Surveyors [RICS], 2018).

The pre-contract stage and the post-contract stage are the two primary stages into which quantity surveyors' roles and responsibilities can be broadly separated in contemporary construction practice (Newton Fisher, 2024). Quantity surveyors are primarily involved in feasibility studies, cost planning, estimating, Bills of Quantities preparation, tender documentation, and advising on appropriate procurement and contract strategies during the pre-contract stage. The post-contract phase, on the other hand, concentrates on contract administration, work valuation, cost control, variation management, and final account preparation (Stonehaven, 2025). Strong technical expertise, sound judgment, and efficient collaboration with a range of stakeholders, including clients, consultants, and contractors, are necessary for both phases (Royal Institution of Chartered Surveyors [RICS], 2022).

In order to analyse and contrast quantity surveying procedures at the pre-contract and post-contract phases, this report offers a case study of a particular construction industry organization. With a focus on the procurement system and contract type used, the study seeks to illustrate the functions and duties of quantity surveyors within the selected organization. Additionally, the financial and financial ramifications of various contract and procurement strategies are examined to show how they affect project cost control and financial performance. The report aims to improve comprehension of the professional roles of quantity surveyors in the construction industry by connecting theoretical knowledge with actual industry practice through this case study.

Summaries:

“The vital role that quantity surveying plays in guaranteeing project value and financial control throughout the construction lifecycle is explained in this section. It describes the different duties between the pre-contract such as feasibility, estimating, tendering and post-contract stages such as administration, valuations and final accounts. Lastly, it outlines the goal of the report, which is to use a case study analysis of procurement strategies and their effects on project success to bridge theory and practice.”

1.1 Background of Quantity Surveying

Within the construction industry, quantity surveying is a professional discipline that focuses on managing contracts and construction costs over the course of a project. A quantity surveyor is in charge of making sure that projects are finished within the allocated budget while upholding the necessary standards of quality and contractual compliance. Due to the growing complexity, scale, and financial risks associated with contemporary construction projects, quantity surveying has become more crucial (Ashworth, A., & Hogg, K. (2007)).

Quantity surveying is a professional discipline in the construction industry that focuses on controlling contracts and construction costs throughout a project. A quantity surveyor is responsible for ensuring that projects are completed within the budget allotted while maintaining the required standards of quality and contractual compliance. Quantity surveying has become increasingly important because modern construction projects are becoming more complex, large-scale, and financially risky (Cunningham, T. (2014)).

Both the public and private construction sectors in Malaysia depend heavily on quantity surveying. Quantity surveyors work for a variety of companies, including government agencies, consulting firms, developers, and contractors. The role of the quantity surveyor has changed to satisfy industry demands while guaranteeing transparency, cost effectiveness, and value for money in construction projects with the introduction of various contract forms and procurement systems (Zakaria, N., et al. (2012)).

Summaries:

“In order to reduce financial risk in complicated projects, quantity surveying is an essential construction discipline that focuses on cost and contract management. This role is crucial in Malaysia's public and private sectors, changing to guarantee value for money and transparency in the face of shifting industry demands.”

1.2 Overview of the Selected Organisation

“This case study is based on a traditional procurement residential project in Selangor with an estimated contract sum of RM12 million where the quantity surveying services were provided by our FKA QS Consultant” (Project Documentation, 2024).

This report is based on a construction project in Malaysia that used traditional procurement. The quantity surveyor was hired as a consultant to handle contract and cost issues both before and after the contract was signed. An organization involved in quantity surveying that works in Malaysia's construction sector has been chosen for this case study. The company offers expert services in the areas of financial control, contract administration, and cost management for building projects. It works on projects in a number of industries, including infrastructure, business, and residential development (Board of Quantity Surveyors Malaysia, 2020).

In both the pre-contract and post-contract stages, the organization is crucial. Its duties include creating cost estimates, bills of quantities, tender documentation, and counselling clients on appropriate contract forms and procurement techniques during the pre-contract phase. Before construction starts, these services help clients comprehend the project's financial implications (Ashworth et al., 2013).

The organization is in charge of cost monitoring, interim valuations, variation assessments, and final account preparation during the post-contract phase (Seeley & Winfield, 1999). The organization helps reduce financial disputes and guarantees that projects are finished within the agreed-upon budget and contractual framework by managing costs and contractual issues throughout the construction period. The choice of this company offers a useful illustration of how quantity surveying concepts are implemented in actual building projects.

Summaries:

This case study examines a RM12 million residential project in Selangor that used traditional procurement and FKA QS Consultant's services (Project Documentation, 2024). From pre-contract cost estimation (Ashworth et al., 2013) to post-contract valuations and final accounts (Seeley & Winfield, 1999), the company oversaw the project's financial lifecycle. The company reduced disagreements and made sure the project stayed within budget by closely monitoring expenses and contracts in accordance with industry standards (Board of Quantity Surveyors Malaysia, 2020).

1.3 Objectives of the Study

The study's goals are to improve comprehension of quantity surveying practice in actual construction scenarios and to conform to the requirements of the Introduction to Quantity Surveying course (SBEQ1343).

The objectives of this study are:

1. to list and describe the pre-contract and post-contract duties and responsibilities of a quantity surveyor.
2. to investigate how the chosen organization manages construction projects using quantity surveying techniques.
3. To examine how the chosen organization's contract types and procurement systems affect project costs.
4. To assess how quantity surveying procedures differ and are similar in the pre-contract and post-contract stages.
5. To improve students' comprehension of practical case study applications in quantity surveying.

By accomplishing these goals, the study hopes to close the gap between theoretical understanding and practical quantity surveying in the building sector.

Summaries:

This study seeks to reconcile theoretical knowledge with practical applications in Quantity Surveying, in accordance with the requirements of course SBEQ1343. The main goals are to look at the company's specific responsibilities before and after the contract, see how procurement systems affect cost management, and compare how procedures change from one stage of a project to the next. Ultimately, the research aims to improve students' understanding of practical industry practices and construction management methodologies.

2.0 Overview of Quantity Surveying Practice

In order to guarantee budget compliance and value for money from start to finish, quantity surveying practice includes the thorough management of financial and contractual aspects in construction projects (Ashworth et al., 2013). Quantity surveyors work with the project team as central cost managers to reduce risks and manage costs in two different stages: the pre-contract stage, which includes cost planning and tendering, and the post-contract stage, which concentrates on valuations, variations, and final accounts (Seeley & Winfield, 1999). The position now requires sophisticated analytical and communication skills to successfully navigate intricate procurement systems as the industry changes, going beyond simple measurement (Zakaria et al., 2012).

2.1 Roles of Quantity Surveyor in Construction Industry

Key Function	Specific Responsibilities & Citations
Cost Planning & Financial Management	comprises giving precise cost guidance and creating project budgets, cost estimates, and cost plans to guarantee financial viability (Ashworth et al., 2013).
Procurement & Tendering	entails creating tender documents, assessing contractor offers, and suggesting qualified contractors based on capability and cost (Rashid et al., 2006).
Contract Administration	focuses on preparing final accounts during the construction phase, valuing finished work, evaluating variations, and offering claims advice (Seeley & Winfield, 1999).
Overall Project Delivery	oversees the contractual and financial framework during the pre- and post-contract phases to guarantee timely and cost-effective delivery (Zakaria et al., 2012).

Summaries:

Financial Viability: Ensuring the project is financially feasible by offering accurate cost guidance and budgets.

Managing tender documents and assessing contractor offers to guarantee equitable selection is known as procurement.

Contract administration: Assessing completed work and controlling deviations to reduce construction-related disputes.

Project Delivery: Managing the complete financial structure to ensure prompt and economical delivery.

2.2 Importance of Pre-Contract & Post-Contract Stages

Because they have a significant impact on a construction project's financial success, the pre-contract and post-contract phases are essential components of quantity surveying practice. Because it concentrates on early cost planning, feasibility studies (Ashworth et al., 2013), and choosing a suitable procurement strategy (Rashid et al., 2006), the pre-contract stage is crucial. Project costs and risk allocation may be impacted in the long run by decisions made at this stage, such as determining the project budget, contract type, and procurement method.

Before construction begins, uncertainty and financial risk can be decreased with effective pre-contract management. Quantity surveyors make sure that both clients and contractors comprehend the project scope and anticipated costs by creating precise cost estimates, bills of quantities, and tender documents (Ashworth et al., 2013). Later in the project, this helps avoid miscommunications and financial problems.

Because it entails managing and keeping an eye on construction-related expenses, the post-contract stage is just as crucial. Quantity surveyors are now in charge of preparing the final account, handling variations and claims, and performing interim valuations (Seeley & Winfield, 1999). Good post-contract management reduces disagreements, ensures equitable financial settlement upon project completion, and keeps the project within the allocated budget.

A comprehensive framework for cost and contract management during the construction process is provided by the pre-contract and post-contract phases taken together (Zakaria et al., 2012). These duties are primarily performed by the quantity surveyor in the chosen case study both before and after the contract.

Summaries:

Pre-Contract Stage (Risk Mitigation): This stage combines strategic procurement selection (Rashid et al., 2006) with early cost planning and feasibility (Ashworth et al., 2013). This stage establishes a clear scope and lowers future financial risks by producing accurate estimates and tender documents (Ashworth et al., 2013).

Post-Contract Stage (Cost Control): This stage uses final accounts, variations, and interim valuations to control ongoing construction costs (Seeley & Winfield, 1999). This ongoing oversight reduces disagreements and guarantees that the project stays within the budget.

3.0 Pre-Contract Quantity Surveying Practice

Pre-contract quantity surveying practice includes all the cost and procurement services provided before the building contract is signed. The goal is to ensure the project is financially viable and well-structured before construction begins (Measure Manage, 2025). This phase starts at the earliest planning stage and continues until the contract is awarded. The quantity surveyor collaborates closely with the client and design team to set budgets, manage cost risks, and prepare the project for tendering and contract signing (Rory Connolly QS, 2024).

Key pre-contract duties include feasibility studies, where the quantity surveyor assesses whether the proposed project can fit within the client's budget. Cost planning involves developing and updating detailed cost plans as the design progresses (Rory Connolly QS, 2024). The quantity surveyor also advises on the procurement strategy, prepares bills of quantities and tender documents, manages the tender process, and helps with tender evaluation and contract negotiations. This ensures that the client can select a contractor on a fair and competitive basis (Pryco Global, 2024).

Summaries:

Pre-contract quantity surveying involves all the planning, costing, and procurement activities that happen before a building contract is signed. The goal is to help ensure the project is affordable and well-organised before construction begins (Measure Manage, 2025). This work often includes assessing if the project is feasible, developing cost plans, giving advice on procurement options, preparing bills of quantities and tender documents, and supporting the client with tender evaluation and contract negotiations. These steps help the client appoint the right contractor fairly and competitively (Rory Connolly QS, 2024).

3.1 Feasibility Study and Cost Planning

In the feasibility stage, the quantity surveyor assesses whether the proposed project can be delivered within the client's budget and is financially viable (Measure Manage, 2025). The quantity surveyor prepares preliminary cost estimates based on scope, location, and market prices, then advises the client whether the project should proceed or be adjusted (Measure Manage, 2025).

After the feasibility study, the quantity surveyor develops a cost plan that breaks down the budget into main cost elements such as materials, labour, and contingencies (Quantity Surveying Coach, 2024). Cost planning allows the quantity surveyor to check design decisions against the budget, suggest cheaper alternatives, and support value engineering so that the project stays within the approved cost limit without reducing required quality (Quantity Surveying Coach, 2024).

In a residential project in Selangor using traditional procurement, the quantity surveyor would first carry out a feasibility study to confirm that the project is financially viable within the approved budget. Then, through structured cost planning, the quantity surveyor would help reduce the risk of cost overruns and support the client's decisions so the project can proceed on a financially sustainable basis (Measure Manage, 2025).

Summaries:

Quantity surveyors use feasibility studies and cost planning to see if the project is affordable. Quantity surveyors also use these tools to keep the budget under control before construction begins (Measure Manage, 2025). In the feasibility stage quantity surveyors estimate the costs. Quantity surveyors advise the client whether the project can go ahead within the approved budget (Measure Manage, 2025; ST1 Group, 2025). In cost planning, they break down the budget into elements, compare design options, and suggest value engineering so the design stays within budget without reducing required quality (Quantity Surveying Coach, 2024).

3.2 Estimating and Budget Control

Quantity surveyors are responsible for preparing cost estimates and controlling the project budget so that construction work can be completed within the client's approved cost limit (Mastt, 2024). Through estimating and ongoing budget control, they help clients manage financial risk and avoid cost overruns on construction projects (Rory Connolly QS, 2025).

In estimating, the quantity surveyor measures the required quantities of materials, labour, and other resources from drawings and specifications and applies current market rates to produce realistic preliminary and detailed cost estimates (Kreo, 2023). These estimates form the basis of the project budget and allow the client to understand likely costs before committing to construction (Cost Logic, 2025).

In budget control, the quantity surveyor uses the approved estimate as a cost target and then monitors design changes, scope variations, and updated price information against this budget (Newton Fisher, 2025). When costs start to exceed the budget, the quantity surveyor highlights the issue and recommends cost-saving options or design adjustments to keep the project financially on track (Cost Logic, 2025).

For our FKA QS case study, estimating and budget control describe how FKA QS prepared realistic cost estimates for the residential project in Selangor and then used these estimates as a basis for monitoring and controlling the budget under the traditional procurement method. By checking design decisions against the approved cost limit and advising on cost-effective alternatives when necessary, FKA QS helped ensure that the project remained financially viable for the client (Mastt, 2024).

Summaries:

Quantity surveyors estimate project costs and control the budget so the work stays within the client's approved cost limit (Mastt, 2024). They calculate quantities and prices to produce realistic estimates, then monitor costs against the budget and suggest savings if costs begin to rise (Rory Connolly QS, 2025).

3.3 Procurement Method Used

Quantity surveyors play a key role in traditional procurement by managing cost and contract matters from design through to construction (Quantity Surveying Hub, 2023). Their work helps the client obtain a well-defined design, competitive tenders, and a controlled final cost under the design–bid–build route (Oktra, 2025).

In traditional procurement, the quantity surveyor advises on procurement strategy, prepares bills of quantities, and produces pre-tender estimates to check that the design is within budget before tenders are invited (Constructive Studio, 2025). The quantity surveyor then manages the tender process, evaluates contractors' bids, and makes recommendations so the client can select a suitable contractor on a fair and transparent basis (Quantity Surveying Hub, 2023).

Once the contract is awarded, the quantity surveyor administers the contract financially by valuing work done, assessing variations, and preparing interim and final accounts to ensure the project is delivered within the agreed contract sum (Constructive Studio, 2025). Throughout, the quantity surveyor provides cost control and commercial advice, helping to manage financial risks and support informed decision-making (Quantity Surveying Hub, 2023).

In our FKA QS Selangor residential case study, the FKA QS acted as the cost and contract specialist under the traditional procurement method by advising on procurement, preparing cost documents for tender, evaluating bids, and then helping the client control the contract price during construction so the project remained within the approved budget.

Summaries:

Quantity surveyors estimate project costs and control the budget, so the work stays within the client's approved cost limit (Mastt, 2024). They calculate quantities and prices to produce realistic estimates, then monitor costs against the budget and suggest savings if costs begin to rise (Newton Fisher, 2025).

3.4 Tendering Process

Quantity surveyors also make sure that the client can pick a contractor fairly and with confidence (Quantity Surveying Coach, 2024). During preparation the quantity surveyor puts together the bill of quantities and other tender documents so that all bidders price the information. The quantity surveyor aims to lower confusion and disputes (Soben, 2024). The quantity surveyor then manages the tender period by issuing documents, responding to contractors' queries, and ensuring that all addenda or clarifications are communicated to every bidder to maintain a level playing field (Quantity Surveying Coach, 2024).

I have seen the quantity surveyor check each bid, for errors, requirements and missing items when tenders are returned. I have seen the quantity surveyor carry out a tender analysis to compare prices and spot any rates or risks (Mastt, 2025). I have seen the quantity surveyor prepare a tender report. The tender report summarizes the bid comparison. I have seen the tender report give advice on which contractor offers the value, for money, not the lowest price (Pride Road Architects, 2021). The quantity surveyor may also assist with post-tender negotiations and adjustments before the contract sum is agreed and the building contract is executed (HC Services, 2023).

In the FKA QS Selangor residential case study, the FKA QS acted as the cost and tendering advisor by preparing the bill of quantities and tender documents, managing the tender process, analysing the contractors' bids, and recommending a suitable contractor for appointment under the traditional procurement method, helping the client achieve a competitive and well-controlled contract price.

Summaries:

Quantity surveyors manage the tendering process by preparing clear tender documents, issuing them to contractors, and then analysing returned bids so that all prices are based on the same scope of work (Quantity Surveying Coach, 2024). They check tenders for errors and qualifications, compare prices, and prepare a tender report recommending the contractor that offers the best value for money, sometimes after post-tender negotiations (Mastt, 2025)

3.5 Contract Documentation

Quantity surveyors are closely involved in contract documentation, helping to turn the agreed tender into a clear, enforceable contract between the client and the contractor (Mastt, 2025). Their work ensures that what was priced and accepted at tender is properly recorded in written documents so that both parties understand their obligations and the basis of payment (ProcurePro, 2025).

In general, contract documentation includes documents such as the form of contract, conditions of contract, specifications, drawings, the bill of quantities, and the signed form of tender and letter of acceptance (Asif, 2025). The quantity surveyor assists in compiling and checking these documents, making sure that the final contract set is consistent, complete, and reflects all agreed clarifications and post-tender adjustments (Soben, 2024).

For our FKA QS Selangor residential case study, the FKA QS helped prepare and assemble the contract documents after tender, using the accepted bill of quantities, tender form, and conditions of contract, so that the client and the selected contractor had a clear, coordinated contract package under the traditional procurement method.

Summaries:

Quantity surveyors help turn the accepted tender into a clear written contract by assembling and checking documents like the form of contract, conditions, drawings, and bill of quantities (Mastt, 2025). They make sure the contract documents are complete and consistent so that both client and contractor clearly understand their obligations and the agreed basis for payment (Soben, 2024).

4.0 Post-Contract Quantity Surveying Practice

Post-contract quantity surveying practice refers to professional services carried out after the construction contract has been awarded and site work has commenced. At this stage, the quantity surveyor focuses on cost control, contract administration, financial management, and settlement of accounts until project completion. The main objective is to ensure that the project is completed following the contract requirement while maintaining effective control over project costs and payments (QSC Engineer, 2023).

Under a traditional procurement system, the quantity surveyor acts as an independent consultant representing the client's financial interests. Post-contract duties include interim valuations, assessment of variations, cost monitoring and preparation of the final account. These responsibilities are critical in preventing cost overruns and contractual disputes during and after construction (Designing Buildings Wiki, 2023).

For the Selangor residential project with an estimated contract sum of RM12 million, FKA QS Consultant played a key role in managing post-contract activities. They acted as the link between the client and contractor by monitoring work progress, verifying claims and ensuring that all payments were made according to the contract. Their professional involvement helped maintain financial control and contract compliance along the construction period.

Effective post-contract management by the quantity surveyor is vital to ensure fair payment, reduce disputes and achieve a smooth project completion. Through careful monitoring and financial oversight, FKA QS Consultant ensured that the project remained on track in terms of cost and contractual obligations.

Summaries:

Post-contract quantity surveying work is important to make sure the project follows the contract and costs are controlled. The quantity surveyor helps manage money and contracts so the project can finish smoothly and fairly.

4.1 Contract Administration

Contract administration is a major responsibility of the quantity surveyor during the post-contract stage. It involves managing and enforcing the terms and conditions of the construction contract to ensure that all parties fulfil their contractual obligations. The quantity surveyor provides advice on matters such as payments, variations, claims, and extensions of time, based on the rules set out in the contract (MedCrave, 2017).

In the Selangor residential project, FKA QS Consultant played an important role in advising the client on contractual matters arising during construction. When unforeseen site conditions required changes to the original scope of work, the quantity surveyor assessed the validity of variation requests and advised on their cost implications. This ensured that only justified variations were approved and that the client was fully informed before any decision was made (Designing Buildings Wiki, 2022).

The quantity surveyor also maintained proper contract records, including site instructions, correspondence, progress reports, and financial documentation. These records are essential for transparency and provide supporting evidence in the event of claims or disputes (IRE Journals, 2017). In cases where delays occurred, FKA QS Consultant advised the client on extension of time matters in accordance with the contract conditions, helping to resolve issues fairly without affecting project progress.

Through effective contract administration, FKA QS Consultant reduced misunderstandings, avoided unnecessary disputes, and ensured that the project was carried out within the agreed contractual framework.

Summaries:

Good contract administration keeps the project organised and reduces disputes. The quantity surveyor gives advice on variations, claims, and delays to protect both the client and the contractor.

4.2 Valuation of Work and Progress Payments

Valuation of work is a key post-contract function performed by the quantity surveyor to determine the value of construction work completed at regular intervals. Interim valuations are prepared based on site measurements, progress reports, and contract rates. These will form the basis for progress payments to the contractor (RICS, 2015).

Timely interim valuations are important because they help maintain the contractor's cash flow and ensure continuous progress on site. Regular payments motivate the contractor to continue work efficiently and reduce the risk of delays caused by financial difficulties (Designing Buildings Wiki, 2022). At the same time, accurate valuations protect the client by ensuring that payments reflect the actual value of work completed, to prevent over payment (Proventus, 2022).

In the Selangor residential project, FKA QS Consultant carefully verified the contractor's claims before certifying interim payments. They ensured that all valuations complied with the contract conditions and that retention sums were properly deducted. This provided financial fairness to both parties and allowed the client to clearly monitor the project's financial status throughout construction (RICS, 2015).

By managing interim valuations efficiently, FKA QS Consultant ensured the project continue smoothly, protected the client's financial interests and made a transparent payment procedure.

Summaries:

Valuing work correctly ensures contractors are paid fairly and on time. This helps the project keep moving while the client knows exactly how much has been spent.

4.3 Variation Orders and Cost Control

Variations are changes to the original contract scope and may arise due to design changes, unforeseen site conditions or changes in client requirements. Managing variations is an important responsibility of the quantity surveyor during the post-contract stage. Each variation must be assessed and valued using contractual rates or fair valuation principles to determine its cost impact (Springer, 1982).

Cost control is closely linked to variation management. The quantity surveyor continuously monitors project expenditure against the approved budget and advises the client on the financial implications of proposed changes. Effective cost control helps prevent excessive cost increases and ensures that the project remains financially viable (Bauwise, 2024).

In the Selangor residential project, several design changes during construction resulted in variation orders. FKA QS Consultant assessed these variations, prepared cost estimates for client approval, and monitored their cumulative impact on the overall project budget. This allowed the client to make informed decisions and understand how variations affected the final project cost.

Through systematic variation assessment and continuous cost monitoring, FKA QS Consultant played a key role in minimising unnecessary expenditure and maintaining cost certainty along the construction period.

Summaries:

Managing variations and controlling costs helps prevent the project from going over budget. The quantity surveyor checks all changes and costs so the client can make smart decisions.

4.4 Final Account and Project Closure

The final account represents the total agreed cost of the construction project upon completion. It includes all contract sums, variations, provisional sums and claims that occurred during the construction period. The quantity surveyor is responsible for preparing, negotiating and finalising the final account to achieve a fair financial settlement between the client and contractor (HKIS, 2015). For the Selangor residential project, FKA QS Consultant assisted both parties in resolving outstanding financial matters and ensuring that all approved works were properly accounted. This helped the client to avoid over payment and ensured that the contractor received correct payment for work. Proper final account preparation also reduced the risk of post-completion disputes (Zakaria et al., 2013). Project closure involves issuing final certificates and completing all contractual and financial obligations. A well-agreed final account provides financial clarity and supports a smooth handover process. Through effective final account management, FKA QS Consultant contributed to a fair, transparent and orderly closure of the construction project.

Summaries:

A clear and agreed final account makes sure all payments are correct and fair. The quantity surveyor helps finish the project properly and reduces the chance of disputes after completion.

5.0 Procurement and Contract Analysis

Choosing the correct procurement and contracts is a critical strategic decision that affects the project's cost certainty, risk allocation, time performance, and the quantity surveyor's responsibilities throughout its lifecycle (Panakaduwa *et al.*, 2025). The type of contract and procurement method applied affect not only how the project is carried out, but also how the client and contractor share financial risks and how effective expenses can be planned, tracked, and controlled. In this case study, the Selangor residential project used a traditional procurement method together with a lump sum contract. This chapter reviews the reliability of these options by looking at their impact on quantity surveying practice throughout both the pre-contract and post-contract stages. The analysis also examines how other types of contracts and procurement processes could have impacted the project's expenses, risks, and financial outcomes.

Summary:

The use of traditional procurement and a lump sum contract had a major impact on cost certainty, risk allocation, and the scope of quantity surveying tasks throughout the project. These strategic decisions influenced both pre-contracts planning efficiency and post-contract cost control performance.

5.1 Procurement System Applied in the Case Study

This project uses a traditional procurement system. This is a usual method of purchasing in the Malaysian construction field, specifically for residential buildings which need cost certainty and design control. In this method, the client hires consultants such as engineers, architects, and quantity surveyors to finalize the design before bidding on the construction projects and awarding it to a contractor (A. Okereke *et al.*, 2022).

Traditional procurement allows people to get highly involved before the contract is signed, which is important for quantity surveying. This is because generating thorough Bills of Quantities, precise cost estimates, and high-quality tender documents becomes easy by obtaining all the design information (A. Okereke *et al.*, 2022). This strategy contributed to the project's cost certainty before signing the contract. This is proven by the tender amount of RM11.95 million, which was quite similar to the pre-tender estimate.

Compared to other methods of purchasing, such as Design and Build, traditional procurement gives clients more control over design quality and makes expenses more transparent. Through Design and Build, the contractor is responsible for the design. This may reduce design adjustments, but risk-related costs normally raise tender prices (Supardi, 2022). Subsequently, the quantity surveyor was able to better control costs before to the beginning of construction through traditional procurement (Kamoche and Wanyona, 2024). However, it also indicated that the project would take longer to complete and would be more probable to change after the contract was signed because the client decided to change the design (A. Okereke *et al.*, 2022).

In general, traditional procurement was the best option for this project because the client preferred to be guaranteed about the costs and quality. However, how well it performed was highly dependent on the completeness of the pre-contract design information. This highlights the importance of the quantity surveyor throughout the initial stages of planning and documentation.

Summary:

Traditional procurement was ideal for this residential project because it provided good cost transparency and design control, while allowing the quantity surveyor to make major contributions during the pre-contract stage. However, its performance was highly dependent on accurate design information to reduce post-contract variations.

5.2 Type of Contract Used

The project adopted a lump sum contract, which is a popular type of contract in traditional procurement. In this situation, the contractor agrees to carry out the job for a specific payment based on mutually agreed-upon plans, specifications, and Bills of Quantities (Ting-shang, 2010).

The client was assured of the costs because the contract amount was determined before construction began. This significantly decreased the client's exposure to market price changes, such as the 12% increase in steel reinforcement pricing that occurred between the first estimate and tender submission. If the contract was re-measurement or cost-reimbursable, the adjustment might have cost the client an additional RM150,000 to RM200,000.

However, for a lump sum contract to turn out effective, the paperwork supporting it must be extremely accurate and thorough. In this project, lack of design information led the client to make multiple changes during construction, increasing the project's total cost by around 3.5%. The lump sum contract restricted flexibility while protecting the client from unexpected rising costs. In contrast, a re-measurement contract was easier to adjust the scope of work but made it more difficult to determine how much the job would cost (Mikowati and Gultom, 2023).

From the perspective of quantity surveying, the lump sum contract required strict control over adjustments made after the contract was signed in order to protect the client's financial interests, and thus gave a stronger focus on doing careful planning before contract signing. The contract system was suitable for the project's objectives, as long as proper cost and control of changes were followed during construction.

Summary:

The lump sum contract provided great cost certainty while protecting the client from market pricing changes, especially material cost risings. However, its effectiveness was dependent on accurate pre-contract paperwork and thorough post-contract variation control by the quantity surveyor.

5.3 Comparison of Pre- and Post-Contract Practices

Quantity surveying procedures have significantly different aims, roles, and risks during the pre-contract and post-contract stages. The differences are summarised in Table 5.1.

Aspect	Pre-Contract Stage	Post-Contract Stage
Primary Objective	Cost planning and risk prevention	Cost control and financial administration
Main QS Activities	Estimating, cost planning, tendering, procurement advice	Interim valuations, variation assessment, final account
Cost Basis	Forecasted and estimated costs	Actual measured and valued work
Risk Management	Risk identification and mitigation	Risk control and dispute resolution
Impact of Procurement	Enables competitive tender pricing	Determines variation management complexity
Impact of Contract Type	Requires accurate documentation	Limits cost exposure but reduces flexibility

Table 5.1 Differences between Pre-contract and Post-contract Stages (Mu *et al.*, 2023; Silva *et al.*, 2025)

Summary:

Pre-contract quantity surveying focuses on cost planning and risk mitigation, whereas post-contract practice focuses cost control and financial management. Both stages are closely connected, since effective pre-contract planning decreases expenses and disagreements during post-contract operation.

6.0 Monetary and Cost Implications

The financial and expense factors are the most important elements of quantity surveying since they directly influence the economic success of a construction project. Quantity surveyors serve as financial protectors, making certain that projects stay feasible through precise cost estimation and rigorous budget management (Bailey Partnership, n.d.). This case study reveals that the residential project in Selangor had a projected contract amount of RM12 million and was acquired via a conventional method with a fixed-price contract.

6.1 Cost Control Effectiveness

To achieve true success, effective cost management should start at the pre-contract phase. Industry specialists highlight that early actions, like Value Engineering (VE), enable quantity surveyors to discover cost-saving potentials while maintaining quality (Newton Fisher, 2024). The preliminary feasibility assessment for this project was RM12.5 million. Through the use of value engineering methods in the planning stage, the quantity surveyor lowered the estimate to RM12.0 million, resulting in a savings of RM500,000 prior to the tender's release.

In the construction phase, attention turns to overseeing spending and handling "variations" (scope changes). Studies show that ongoing oversight of changes is crucial to avoid budget exceedances (Tan & Suleiman, 2024). In this project, the contractor claimed variations amounting to RM480,000. The quantity surveyor evaluated these claims rigorously based on the contract rates and successfully negotiated the approved sum down to RM420,000, resulting in a savings of RM60,000 for the client. As a result, the final account was settled at RM12.42 million, indicating a variance of merely 3.5%, which is comfortably within the industry's acceptable range

Summary:

This chapter assesses the financial results of the project, backed by industry-leading practices in cost management, risk assessment, and contract approach

Efficient cost management began with Value Engineering during the pre-contract phase to reduce the initial budget, while post-contract oversight concentrated on negotiating variation claims. Both approaches guaranteed that the final account stayed within a 3.5% fluctuation, showcasing financial prudence during the project's duration

6.2 Financial Risks and Cost Impacts

Construction projects face considerable financial hazards, such as varying material costs and unexpected site circumstances. Bauwise (2025) states that instability in market-driven rates significantly contributes to budget failure if not addressed proactively. A major risk in this case study was the fluctuation of material costs; notably, the cost of steel reinforcement rose by 12% throughout the project. Had this risk not been addressed, it would have increased the project cost by around RM300,000. A frequent financial risk is "scope creep," in which client-requested changes increase expenses. In this project, modifications in the design amounted to RM250,000, and unexpected site conditions contributed RM170,000. In the absence of efficient risk management, the overall extra expenses might have surpassed RM700,000. By recognizing these risks promptly and implementing stringent contract clauses, the quantity surveyor reduced the real financial effect to RM420,000, thus safeguarding the client's budget.

Summary:

Financial risk management focused on market price fluctuations and scope expansion, while the quantity surveyor's diligent oversight restricted the real cost effects. Both strategies safeguarded the client from overexposure, lowering possible extra expenses from RM700,000 to RM420,000

6.3 Impact of Procurement and Contract Type on Project Cost

The selection of procurement method and contract category essentially determines the distribution of financial risk between the client and the contractor. This project employed a Lump Sum Contract. A key benefit of this contract type is "cost certainty," because the contractor commits to a set price for a specific scope of work, thus taking on the risk of cost overruns from material costs or inefficiencies (NetSuite, 2021). In this case study, the fixed-price contract effectively protected the client from the previously noted rise in steel prices. Due to the established price, the contractor was required to manage the market volatility. Although lump sum contracts may lack flexibility for design alterations (NetSuite, 2021), they provide better financial safeguards for clearly defined projects. The ultimate cost rise of 3.5% is considerably less than the 8-10% increase that could be anticipated with alternative procurement strategies where scope alterations are more challenging to manage.

Summary:

Financial risk management concentrated on changes in market prices and broadened scope, whereas the quantity surveyor's careful supervision limited the impact on actual costs. Both approaches protected the client from excessive exposure, reducing potential additional costs from RM700,000 to RM420,000.

7.0 Questions, challenges and suggestions

In the process of residential development of RM12 million in Selangor, the FKA QS consultant showed professionalism in managing quantitative measurement responsibilities. However, there are various obstacles in the pre-contract and post-contract stages, which affects the financial performance and operational efficiency of the project. This section reviews the special difficulties encountered in this case study and proposes actionable solutions from the actual experience of the project in conventional procurement methods and fixed-price contract arrangements.

7.1 Problems encountered in the pre-contract stage

7.1.1 Incomplete design information and compressed schedule

In the preparation stage for the bidding of the residential development, some building specifications of individual residential units have not been determined, especially for the decorative materials of the kitchen and bathroom. This shortfall requires the inclusion in the quantity list document of a provisional amount of approximately RM180,000, or 1.5% of the total value of the contract. When these design elements were subsequently finalized during the construction phase, the cost realized exceeded the provisional allocation of about RM45,000, so customers were required to obtain supplementary financing.

The above challenges are exacerbated by the limited time frame allocated to FKA QS consultants to prepare bidding documents, which lasts only four weeks. Professional industry standards usually recommend comparable-scale development within 6-8 weeks. This shortened schedule limits the opportunity for comprehensive quantitative verification procedures and value engineering analysis, which have the potential to determine the possibility of cost optimization of RM100,000-150,000.

Root cause analysis: The urgency of the customer's start of construction has shortened the design and development period and accelerated the bidding preparation process. Therefore, FKA QS consultants have to carry out incomplete design information and insufficient duration for thorough document coordination activities.

Summary:

The challenges before the contract include insufficient design specifications and limited preparation schedule. The four-week preparation period for bidding, plus a temporary amount of RM180,000, resulted in an additional expenditure of RM45,000 during the construction period. The industry recommends adequate preparation in 6-8 weeks, which can promote potential value-saving engineering opportunities worth RM100,000 to RM150,000.

7.1.2 Material price fluctuations

There is a three-month interval between the initial cost estimation stage (with a budget of RM12.5 million) and the bid submission deadline. During this period, the price of rebar rose by about 12% due to the interruption of the regional supply chain. Such market fluctuations may increase project expenditure by RM150,000 to 200,000.

Fortunately, the one-time contract arrangement effectively transfers this financial risk to the contractor's responsibility. Therefore, customers are still isolated from this price fluctuation. The contractor's bid of RM11.95 million indicated that they accepted this exposure through competitive pricing strategies or pre-material procurement practices. This situation shows the fundamental advantage of fixed-price contracts: regardless of the market price fluctuations during the construction period, customers can obtain price certainty.

Summary:

Fluctuations in material prices, especially the 12% increase in the cost of rebar, is an important pre-contract risk factor. However, the one-time contract successfully transferred the potential cost increase of RM150,000 to RM200,000 to the contractor, thus protecting customers from market fluctuations. This shows the risk mitigation benefits inherent in the fixed-price contract arrangement.

7.2 Problems encountered in the post-contract stage

7.2.1 Design changes and valuation disputes initiated by customers

The most substantial post-contract challenge involves a large number of design modification requests made by customers after the start of construction. These changes include upgrading the decoration specifications of public areas and modifications to the unit configuration according to the buyer's preferences. The contractor's initial difference claim totaled RM480,000. Through meticulous verification procedures and negotiation processes, the FKA QS consultant successfully reduced the approved difference amount to RM420,000, thus saving RM60,000 for customers, a reduction of 12.5%.

However, each variant order takes an average processing time of 3-4 weeks, which causes a variety of complications. There are differences on the valuation rates of various works, especially when the direct billing interest rate is proven to be inapplicable. For example, when the customer requests to upgrade the floor materials, especially the Italian marble alternative standard tiles, the rate requirements submitted by the contractor are 40% higher than the current market price. After careful negotiation, the final agreed rate was 18% higher than the original billing rate, saving about RM35,000 compared to the contractor's initial claim.

Summary:

It has been proved that post-contract differential management is challenging. Through diligent negotiation, the initial contractor's claim was RM480,000, reduced to RM420,000, saving RM60,000. Each change takes 3-4 weeks to deal with, causing valuation disputes, especially for projects without direct billing interest rates. The effective negotiation of floor upgrade alone saved RM35,000.

7.2.2 Tight cash flow and processing delays

At the same time, many changes were handled, resulting in a backlog in the temporary valuation process. Several payment certificates were delayed by 2-3 weeks beyond the contract schedule, which caused tension between the customer and the contractor. Contractors are concerned that the impact of cash flow affects their ability to retain optimal workforce resources. The fundamental reason stems from the brief combination of pre-contract preparation, which leads to ambiguous specifications and many post-contract changes. This combination overwhelms the standard valuation process. If the situation deteriorates further, customers may face potential financing costs or compensation claims worth about RM30,000-50,000.

Summary:

Concurrency change processing caused a valuation backlog, resulting in a delay of the payment certificate by more than 2-3 weeks of contract terms. These delays put pressure on contractors' cash flow and customer-contractor relationships. The root cause, combined with insufficient pre-contract preparation and excessive post-contract changes, may make customers face claims of RM30,000-50,000.

7.3 Suggestions for improvement

7.3.1 Strengthen pre-contract practices

To solve the challenges before the contract, the FKA QS consultant should advocate the implementation of the formal design freeze milestone at least 6 weeks before the tender is released. All major design components should be required to be authorized by the customer. The agreement will reduce post-contractual changes by about 40-50%, and it is possible to save RM100,000 to RM125,000. In addition, at least 6-8 weeks should be allocated to prepare bidding documents for projects worth RM10 million to RM15 million. This time frame allows for adequate value engineering exercises, strengthen coordination activities, and minimize document errors. For projects with a significant time gap between cost estimation and bidding release, monthly material price trend monitoring should be carried out. This approach can facilitate the best bidding timing to achieve savings of RM50,000-80,000 by taking advantage of favourable pricing conditions.

Summary:

The proposed pre-contract improvements include the implementation of design freeze 6 weeks before bidding, the allocation of 6-8 weeks of document preparation, and monthly material price monitoring. These measures can reduce changes by 40-50%, saving RM100,000 to RM125,000, and the best bidding time can save an additional RM50,000-80,000.

7.3.2 Strengthen post-contract management

To alleviate the challenges after the contract, a structured variation management framework should be established. This should include pre-approval requirements for pre-job changes and an evaluation schedule of up to 10 days. Such agreements will reduce the variation of disputes by 60-70% and speed up the resolution time from 3-4 weeks to 10-14 days.

Pre-negotiated rate tables should be prepared at the bidding stage for common change categories (such as material upgrades and minor design modifications). This method will minimize disputes similar to floor changes, which take 5 weeks to resolve. For minor changes below RM10,000, a simplified assessment procedure shall be implemented, with a maximum approval period of 3 days. In this project, the total small change was about RM90,000, but disproportionately consumed administrative resources.

Digital cost management technology should also be deployed to speed up the processing of mid-term valuations by 50% and provide real-time cost reporting capabilities. This will shorten the valuation processing time from 3-4 weeks to 1.5-2 weeks and eliminate potential claims worth RM30,000-50,000.

Summary:

The post-contract proposal includes structured change management, including a 10-day assessment schedule, a pre-negotiated rate table, and a simplified procedure for changes under RM10,000. Digital tools can reduce the processing time by 50%, from 3-4 weeks to 1.5-2 weeks, avoiding potential RM30,000-50,000 claims, and reducing controversial changes by 60-70%.

7.3.3 Summary of Potential Improvements

If these recommendations had been implemented in this project, the outcomes could have been:

Aspect	Actual Outcome	With Recommendations	Improvement
Final Account	RM12.42M (3.5% increase)	RM12.25M (2.1% increase)	RM170,000 saving
Total Variations	RM420,000	RM250,000- RM300,000	RM120,000- RM170,000 reduction
Valuation Processing Time	3-4 weeks average	1.5-2 weeks average	Better cash flow

These improvements indicate that the systematic application of best practices can provide substantial financial and operational benefits for consultants and customers of traditional procurement of residential projects.

Summary:

The implementation of all the recommendations could have reduced the final account from RM12.42 million to RM12.25 million, saving RM170,000. The total change may be reduced from RM120,000 to RM170,000, and the processing time will be halved from 3-4 weeks to 1.5-2 weeks. These improvements show that significant financial and operational benefits can be achieved through the systematic application of best practices.

8.0 Conclusion

The case study of the RM12 million Selangor residential project demonstrates the important role of quantitative surveyors in building cost management. FKA QS consultants use the traditional procurement of one-time contracts to manage this development.

The project has achieved remarkable financial results: before saving RM500,000 in contracts through value engineering, RM60,000 in contracts through change management, and preventing a rise in steel prices of RM150,000 to RM200,000. The final account settlement was RM12.42 million, an increase of only 3.5% over the original contract amount. The measurable total income was RM760,000 to RM810,000, far exceeding the consulting fee of RM120,000 to RM180,000, achieving a return on investment of about 4.5-6.8 times. However, the shortened pre-contract schedule led to an excess cost of RM45,000, and the delay in post-contract change processing by 3-4 weeks made the relationship tense. These challenges highlight the importance of adequate preparation time and simplified procedures.

This study confirms that quantitative measurement is a comprehensive cost management service, and the pre-contract and post-contract stages must act as a comprehensive process. Facts have proved that the traditional procurement of one-time contracts is effective for cost control, but less flexible for design changes. Future procurement choices should be consistent with customer characteristics: traditional procurement is suitable for cost-sensitive customers, while design and construction better serve customers who need flexibility.

The implementation of the proposed improvements could save an additional RM120,000 to RM170,000, indicating that continuous improvements have brought tangible financial returns. By implementing enhanced preparation protocols and simplified change management, quantitative surveyors can strengthen their position as important partners for the success of construction projects.

Summary:

The project showed excellent cost management, with a total income of RM760,000 to RM810,000, which is 4.5-6.8 times higher than the consulting fee. The final cost increase of 3.5% is far in line with industry standards. However, hasty preparation and slow variant processing have resulted in avoidable inefficiency. Facts have proved that the comprehensive approach before and after the contract is crucial to success. Although traditional procurement is effective in cost control, the proposed improvements can save an additional RM120,000 to RM170,000. Strategic procurement selection and continuous improvement will further improve the value contribution of quantitative surveyors.

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